ZeroGUI: Automating Online GUI Learning at Zero Human Cost

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Paper version: arXiv 2505.23762v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Offline learning ... heavy reliance on high-quality manual annotations ... and limited adaptability to dynamic and interactive environments.*

Grounding labels, trajectories and task verifiers are all hand-made; static sets miss moving UIs.

Motivation: why online, at zero cost

1. **Annotation cost.** Grounding and trajectory labels do not scale.
2. **GUIs drift.** Elements shift; agents overfit static tasks.
3. **No train split.** OSWorld / AndroidLab ship tests + verifiers only.

Evaluation setup

OSWorld · 369 tasks | AndroidLab · 138 | UI-TARS-7B-DPO base | Aguvis-7B base | screenshot, 15-step cap | 4 runs, online

- **All online** SR; no offline grounding metrics.
- **Not held out:** stage 2 trains on the test tasks.

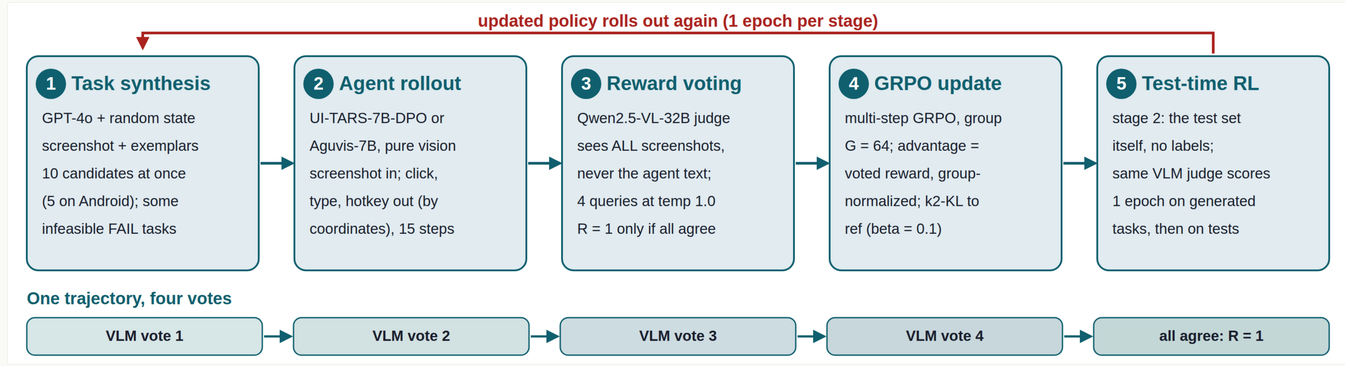
Our read (CUA Radar)

65	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval (OSWorld, AndroidLab); 4-run mean±std; RFT baselines; ablations; judge P/R check; code.
- **Flags:** stage 2 trains on the test set (no held-out env); budget and cost partly matched.
- Pattern K16, learned VLM outcome judge. Authors' numbers, not rerun.

Method: annotation-free online RL

- **Sees / acts:** screenshot in; click / type / hotkey at coordinates out — OSWorld desktop, AndroidLab mobile.
- **Signal:** binary reward = unanimous vote of 4 VLM reads over all screenshots; agent text hidden.
- **Training:** two-stage GRPO — generated tasks first, then test-time training; k2-KL swap for stability.



Our redraw (not the paper's figure): VLM task synthesis, rollout, unanimous VLM reward, multi-step GRPO, then test-time training.

$$\hat{A}^{(i)} = \frac{R^{(i)} - \text{mean}\{R^{(i)}\}_{i=1}^{G}}{\text{std}\{R^{(i)}\}_{i=1}^{G}}$$

GRPO advantage: the voted binary reward, normalized over a group of G = 64 trajectories

Key results: OSWorld SR (%)

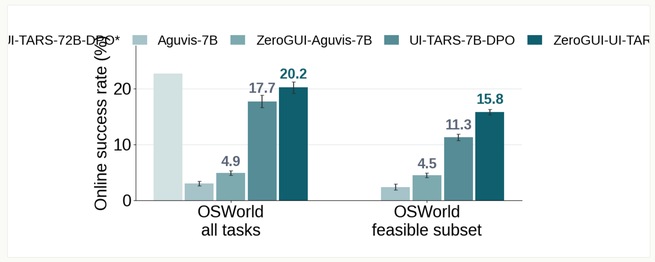


*Table 1 redrawn: mean ± std, 4 runs; * as reported in prior papers.*

+2.5 SR over UI-TARS-7B-DPO (14% rel.); +1.9 over Aguvis-7B (63% rel.)

SO WHAT → *AndroidLab (online): +1.8 SR full set, +2.8 ops subset.*

Ablation: training recipe (Daily SR, %)

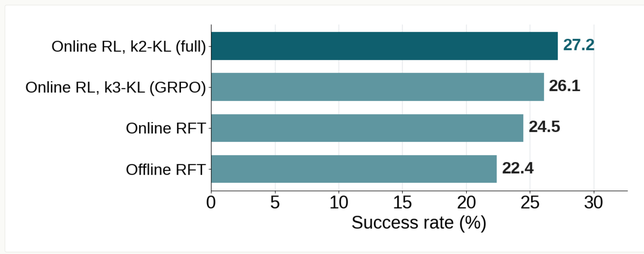


Table 4, redrawn: OSWorld Daily domain (Chrome, Thunderbird, VLC).

SO WHAT → *Online RL (27.2) leads online RFT (24.5) and offline RFT (22.4); k2-KL 27.2 vs k3-KL 26.1. Fig. 6: dropping the KL term (β = 0) or raising it to 1.0 both hurt. The stages split roles: generated-task training lifts pass@4, test-time training lifts all-pass@4.*

Headline Numbers

20.2
OSWorld SR, UI-TARS-7B
+2.5 over the 17.7 base; +4.5 feasible subset

+63% relative SR, Aguvis-7B

+1.8 AndroidLab SR

0 human labels needed

The paper's own controls

VARIANT (OSWORLD DAILY)	SR	JUDGE P / R
Judge: final frame only	23.9	47.5 / 40.4
Judge: reads agent text	24.0	44.3 / 74.5
Judge: no voting	25.6	53.7 / 61.7
Full: all frames + vote	27.2	61.5 / 51.1
Task gen: w/o exemplars	22.3	vs 27.2 full recipe

Checked vs manually labeled trajectories (Table 3b).

Takeaway

GPT-4o tasks + voted VLM rewards + GRPO: +2.5 OSWorld SR, +1.8 AndroidLab, zero labels — test-time training runs on the test set.

judge P 61.5 / R 51.1 | 725 Ubuntu train tasks | 15-step cap, 4 runs | +2.9 Sub-SR, ops subset | code released

Scan to Read

Paper | Code